

FINAL BXP EVIDENCE AUDIT

Date: July 2026

Purpose: Pre-submission quality control for scholarship application evidence package

Auditor: Evidence audit conducted against actual project files

1. What Evidence Exists

Strongest evidence (fully verifiable from this repository):

Item	What it is	Verification
SPEC.md	1,300-line formal protocol specification	Open and readable
reference-server/server.py	1,877-line FastAPI reference implementation	Runnable code
sdk/python/bxp_sdk.py	895-line Python SDK	Runnable code
sdk/typescript/bxp-sdk.ts	TypeScript SDK	Readable code
cli/bxp_cli.py	740+-line CLI tool	Runnable code
datasets/sample_readings.bxp.json	10 validated global city readings	Readable JSON
CHANGELOG.md	Dated development history (v1.0 → v2)	Readable
Zenodo DOI (spec)	Permanent publication record	doi.org/10.5281/zenodo.18906812
Zenodo DOI (implementation)	Permanent publication record	doi.org/10.5281/zenodo.18907003
GitHub repository	Public, timestamped, open-source	github.com/bxpprotocol/bxp-spec
BXP_Evidence_Package/	9-document research evidence package	This directory

2. What Evidence is Strongest

For demonstrating protocol design and systems thinking:

SPEC.md is the strongest single piece of evidence. It is 1,300 lines of formal technical specification covering: a 32-agent schema, a five-stage data pipeline, 20+ REST API endpoints with full schemas, a privacy and security framework, a health risk index with formula, governance model, and regulatory compliance analysis. The specification has a permanent DOI and is publicly available. A reviewer can read it without running any code.

For demonstrating technical implementation:

The reference server (reference-server/server.py) is the strongest implementation evidence. It is a working 1,877-line FastAPI application that implements the BXP specification. A reviewer can clone the repository, install four Python packages, run python server.py, and interact with a live BXP node at <http://localhost:8000>.

For demonstrating intellectual development and iteration:

The CHANGELOG.md combined with the development gap analysis is the strongest evidence of process. The changelog documents the progression from v1.0 (February 15, 2026) through the 50-item systematic gap analysis to v2.0 specification to v2.1 implementation. This shows that the project grew through genuine iteration, not a single flash of insight.

For demonstrating global scope and practical relevance:

The sample dataset (datasets/sample_readings.bxp.json) concretely demonstrates what BXP is for: 10 VALIDATED readings from Accra, Lagos, Delhi, Beijing, London, São Paulo, New York,

Nairobi, Jakarta, and Cairo — all six inhabited continents — in a single file, in the same schema. HRI scores range from 29.1 (New York, MODERATE) to 100.0 (Delhi, HAZARDOUS).

3. What Evidence is Incomplete

Claim	Evidence gap	Honest description
Binary .bxp format	Specified but not implemented	SPEC.md §5.2 gives complete byte-level
Federated node sync	Designed but not implemented	/nodes endpoint exists; peer-to-peer
pip install bxp-sdk	pyproject.toml and setup.py exist	Package has not been published to Py
npm package bxp-sdk	TypeScript SDK exists	Package has not been published to np
Live BXP network	No nodes exist except locally	The reference server is the only imp

4. What Evidence is Missing

What's missing	Why it matters	How to address it (future)
Third-party implementation	Demonstrates that the protocol can b	Open to contributions; engage sensor
Epidemiological review of BXP_HRI	The health risk index is unvalidated	Engage atmospheric scientists; submi
User testing or deployment data	No real-world deployment exists	Partner with a community sensor netw
Formal compliance assessment	GDPR/HIPAA compliance is designed, n	Legal/compliance review by a privacy
BXP Foundation as a legal entity	The governance body is proposed, not	Establish as a registered non-profit

5. What Claims Require External Verification

These claims are made in the project but cannot be verified from the repository alone:

Claim	Verification required	Current basis
"BXP is to air quality data what HTT	Requires adoption to be true at scal	Analogy only; honest about this
"Addresses data fragmentation perman	Requires adoption; cannot be demonst	Design goal; not achieved yet
WHO AQG 2021 threshold alignment	Verifiable against the WHO publicati	WHO thresholds are public and can be
GDPR compliance	Requires legal assessment	Privacy architecture is designed for
BXP_HRI health risk representation	Requires epidemiological review	Weights derived from WHO DALY data;

6. What Should Be Submitted as Scholarship Evidence

Recommended for primary submission:

1. GitHub repository link — reviewers can browse all code, the specification, and the entire history
2. Zenodo DOI for the specification (doi.org/10.5281/zenodo.18906812) — permanent, citable, demonstrates publication
3. BXP_Evidence_Package/01_Independent_Research_Report/ — most comprehensive explanation of the work, the intellectual journey, and honest limitations
4. BXP_Evidence_Package/08_Portfolio_Summary/ — concise one-page summary

Recommended as supporting evidence:

5. SPEC.md — primary technical artefact; demonstrates protocol design depth
6. BXP_Evidence_Package/06_Development_Journey/ — demonstrates intellectual development over time

7. BXP_Evidence_Package/04_Architecture/ — demonstrates systems thinking through diagrams

Recommended as supplementary:

8. Sample dataset (datasets/sample_readings.bxp.json) — concrete, readable evidence of the protocol in action

9. CHANGELOG.md — dated development history

7. What Should NOT Be Submitted

Do not submit or emphasise:

- * Claims that BXP is "widely adopted" or "an established standard" — it is not
- * Claims that BXP_HRI is scientifically validated — it is not
- * The BXP Foundation as an existing organisation — it does not exist
- * Claims of third-party implementations — none exist
- * The binary format as implemented — it is specified, not built

Be prepared to explain these limitations if asked. They are honest limitations of independent early-stage research and should be framed as such.

8. Any Remaining Risks

Risk | Mitigation

Reviewer may ask for external validation	Prepared answer: "The weighting method is based on WHO 2021 and Simcoe 2012, both genuine"
Reviewer may note no external adoption	Prepared answer: "This is an early-stage research project"
Reviewer may note BXP Foundation does not exist	Prepared answer: "The governance model is based on WHO 2021 and Simcoe 2012, both genuine"
Repository may be compared to professional standards	Frame as: "independent undergraduate research project"

9. Quality Control Checklist

- No fabricated claims
 - No unsupported scientific claims
 - No invented citations (two references cited: WHO 2021 and Simcoe 2012, both genuine)
 - No false external validation
 - No contradictions between documents
 - No technical inconsistencies
 - Clear distinction between implemented, specified, proposed, and future work
 - Clear distinction between original contribution and AI-assisted development acknowledged
 - Consistent terminology throughout (BXP_HRI, not "HRI score" in some places and "BXP Health Risk" in others)
 - All file references point to files that exist in the repository
 - Dates are drawn from actual documented sources (CHANGELOG, SPEC.md Origin section)
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Summary Assessment

The BXP evidence package accurately represents an independent research and technical

development project by a single undergraduate-aged creator. The work is genuine, substantive, and publicly verifiable. The specification is thorough and technically serious. The implementation is working and runnable. The documentation is complete and honest.

The strongest story to tell: An unconventional question about why air quality data is fragmented → independent investigation → recognition of a structural interoperability problem → design of a universal open protocol → full technical specification and reference implementation → honest acknowledgement of what remains to be built.

The honest limitation to acknowledge: BXP is an independent research proposal and prototype. It has not been externally adopted, clinically validated, or reviewed by domain experts. These are the natural boundaries of independent research at this stage — and acknowledging them is itself a demonstration of intellectual maturity.

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